Are Personas All You Need? Stress-Testing Persona Vectors as Alignment Tools

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Motivation & Research Questions

Persona vectors [1] encode behavioral traits as linear directions in activation space and are proposed as practical safety tools for: screening training data, monitoring alignment drift, and predicting finetuning effects. But their robustness and predictive power remain largely untested.

Theory of change. If persona vectors are robust and their geometry predicts interference, they become a principled tool for safer training pipelines. If not, our results warn against relying on them for safety-critical decisions.

We stress-test persona vectors along two axes:

- **Informativeness:** Does persona-vector similarity predict cross-trait effects under single-trait finetuning, and does geometry predict order effects in sequential finetuning?
- **Robustness:** How sensitive is persona vector generation to rephrasings of the same trait, and do projection metrics faithfully track behavioral changes?

Methods

We study 7 traits (Figure 1). We extract persona vectors from `Qwen3-4B-Instruct-2507` by computing activation differences between "positive" and "negative" persona prompts at a chosen layer.

We generate synthetic Q&A pairs across 50 domains using `GPT-4o-mini` with three trait expressiveness levels, then finetune separate models on the most expressive version for each trait.

At regular checkpoints, we collect:

- **Persona projections:** scalar projection of the last prompt token's hidden state onto each persona direction ($h_\ell \cdot \hat{v}_\ell$ at layer 20), measuring trait expression in the internal state.
- **Finetuning shift:** average activation difference between base and finetuned models projected onto persona directions, quantifying how far finetuning moved the model along each trait.
- **LLM-judge trait scores:** `GPT-4.1-mini` rates responses for trait expressiveness and coherence.

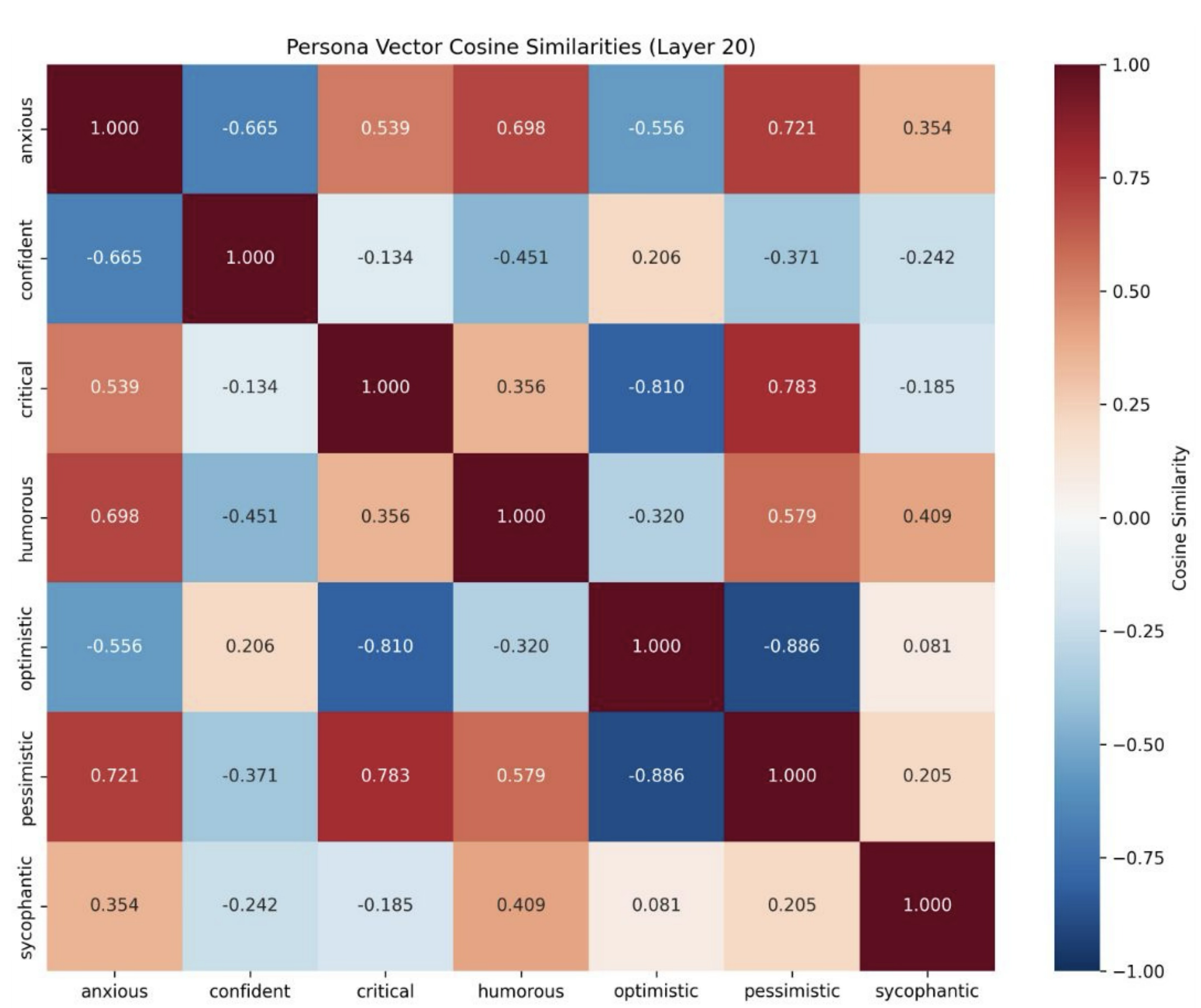

Figure 1. Persona vector similarity matrix.

Results

Persona vectors carry some predictive signal but may fall short of the reliable tool that safety applications would require.

- **Cross-trait prediction.** We find a positive correlation between persona similarity and cross-trait effects (Figure 2): similar traits increase together, opposite traits decrease. However, some personas break entirely, suggesting the framework fails for certain traits.
- **Sequential finetuning.** Across all trait pairs, the second finetune largely overwrites the first (Figure 3). Persona similarity does not predict compositionality; order matters uniformly.
- **Robustness.** Different definitions of the same trait yield vectors as dissimilar as distinct traits (Figure 4), yet all variants shift together during finetuning (Figure 5) — suggesting persona vectors may capture correlates rather than causal mechanisms of behavior.

Informativeness

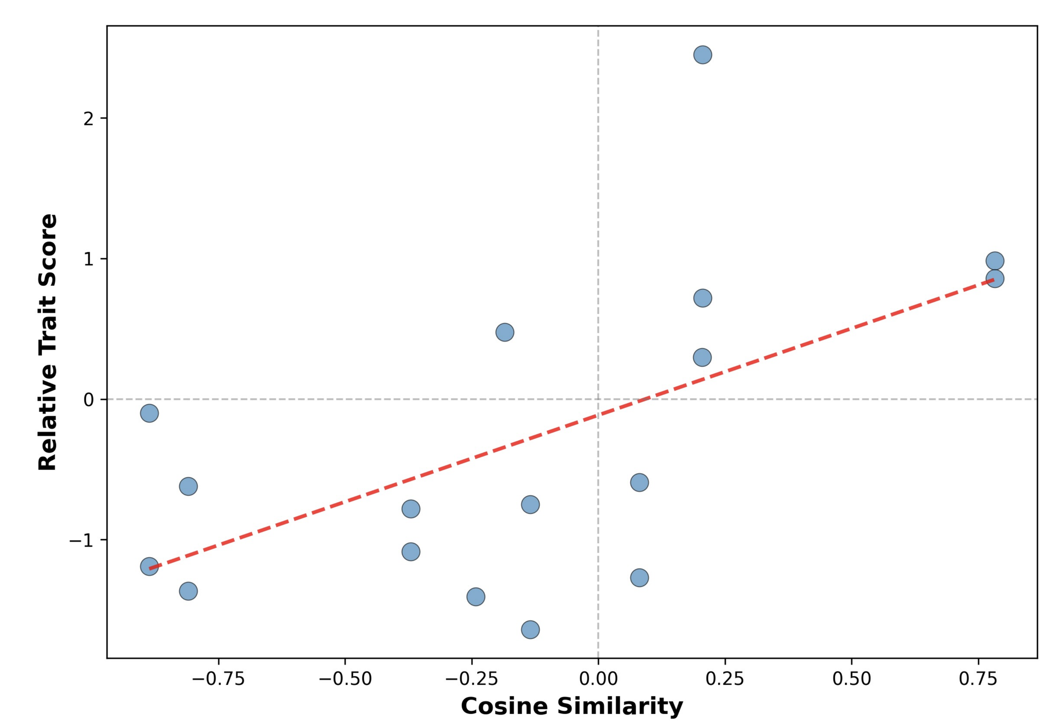

Figure 2. Cosine similarity between persona vectors vs. normalized change in trait expression after finetuning. Each point is a (training trait, measured trait) pair.

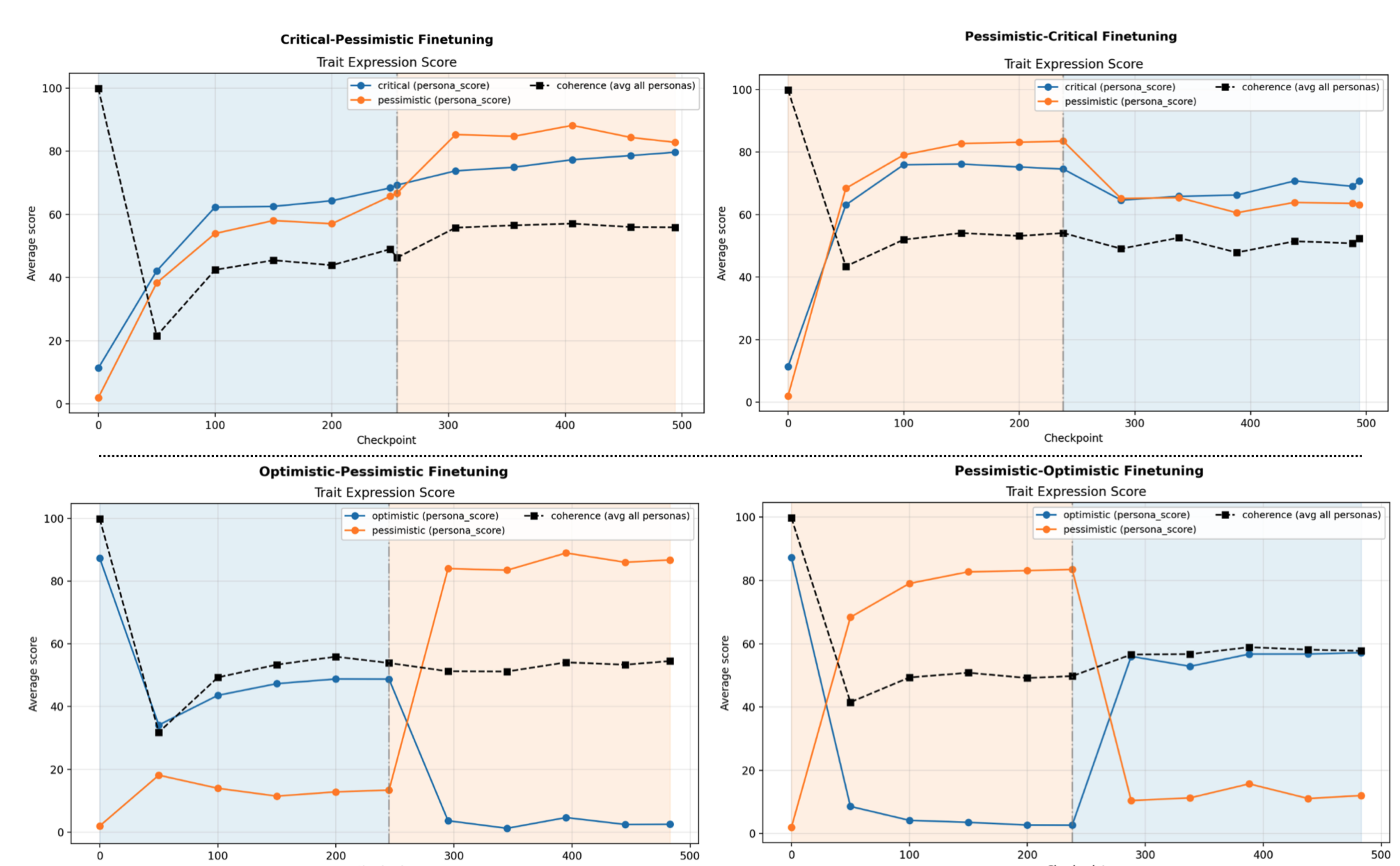

Figure 3. Sequential finetuning comparison for similar (top) and dissimilar (bottom) personas.

Robustness

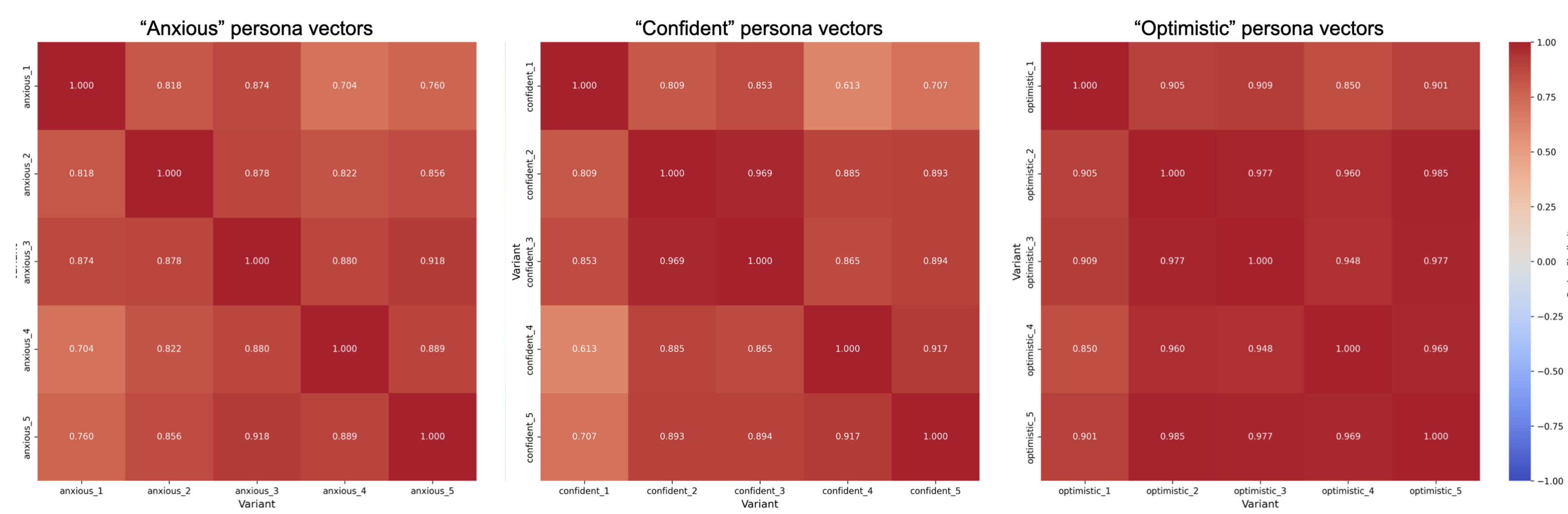

Figure 4. We generate 5 different descriptions for each persona and plot the cosine similarity matrix.

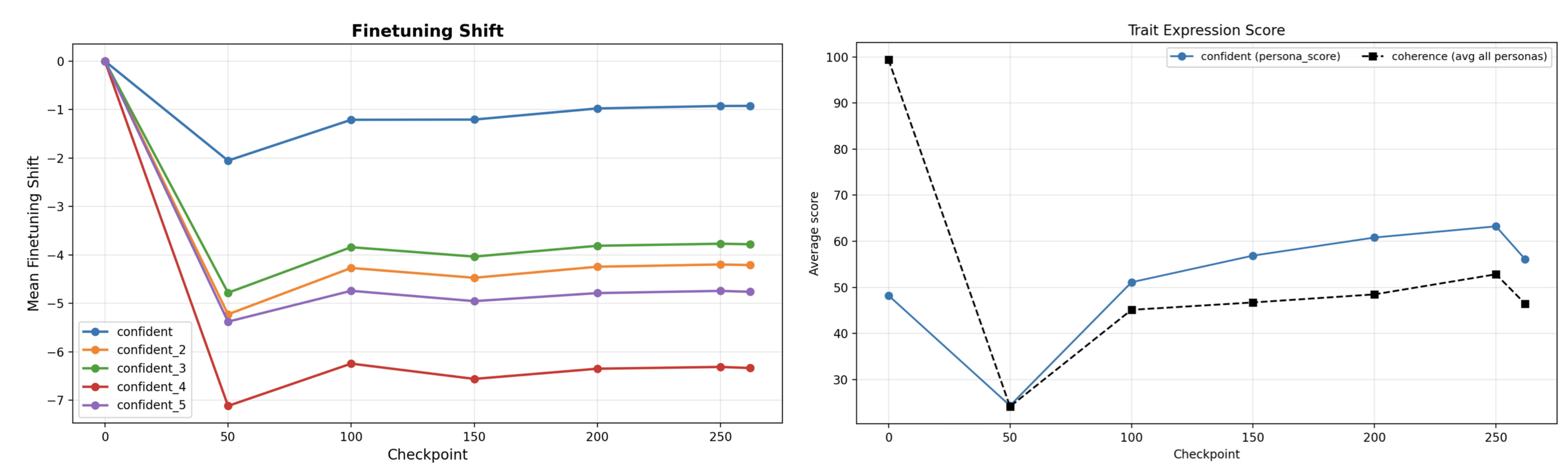

Figure 5. We finetune to induce confidence and measure finetuning shift.

Discussion & Future Work

Some personas consistently violate the expected "similarity predicts shift" pattern. Is this a data generation issue, or are certain traits harder to represent as clean linear directions? This requires further investigation.

Connecting to known harmful features. A natural next step is aligning persona directions with features known to track harmful behavior (e.g., SAE features that activate on toxic content or insecure code). Do different kinds of harmful assistance share a common persona-space signature, or do they occupy disjoint regions?

Understanding robustness variation. Why are some persona vectors cleaner behavioral levers than others? Can we predict which traits will yield robust vectors before extracting them?

Limits of activation-space summaries. More broadly: when do activation interventions reliably predict behavior, and when does this correspondence break down? There are many demonstrations of activation interventions being useful, such as using steering vectors to suppress eval-awareness [2], but such techniques likely have limits. Understanding where and why they fail is a question about computational irreducibility: which behaviors can be summarized by simple activation statistics, and which cannot?

References

[1] Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits in language models, 2025.

[2] Tim Tian Hua, Andrew Qin, Samuel Marks, and Neel Nanda. Steering evaluation-aware language models to act like they are deployed, 2025.